



OBJECT ORIENTED SOFTWARE ENGINEERING

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Advanced relationship in Class Model –

1. Generalization and Inheritance relationships

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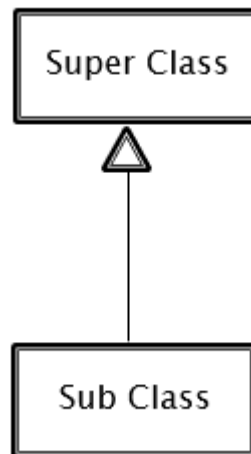
- Relationship between a class (the superclass) and one or more variations of the class (the subclass)
- Simple generalization **organizes classes into a hierarchy**; each subclass has a single immediate superclass. There can be **multiple levels of generalizations**.
- **Generalization, specialization and inheritance** all refer to aspects of the same idea.

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Generalization or Inheritance

UML notation for generalization or inheritance

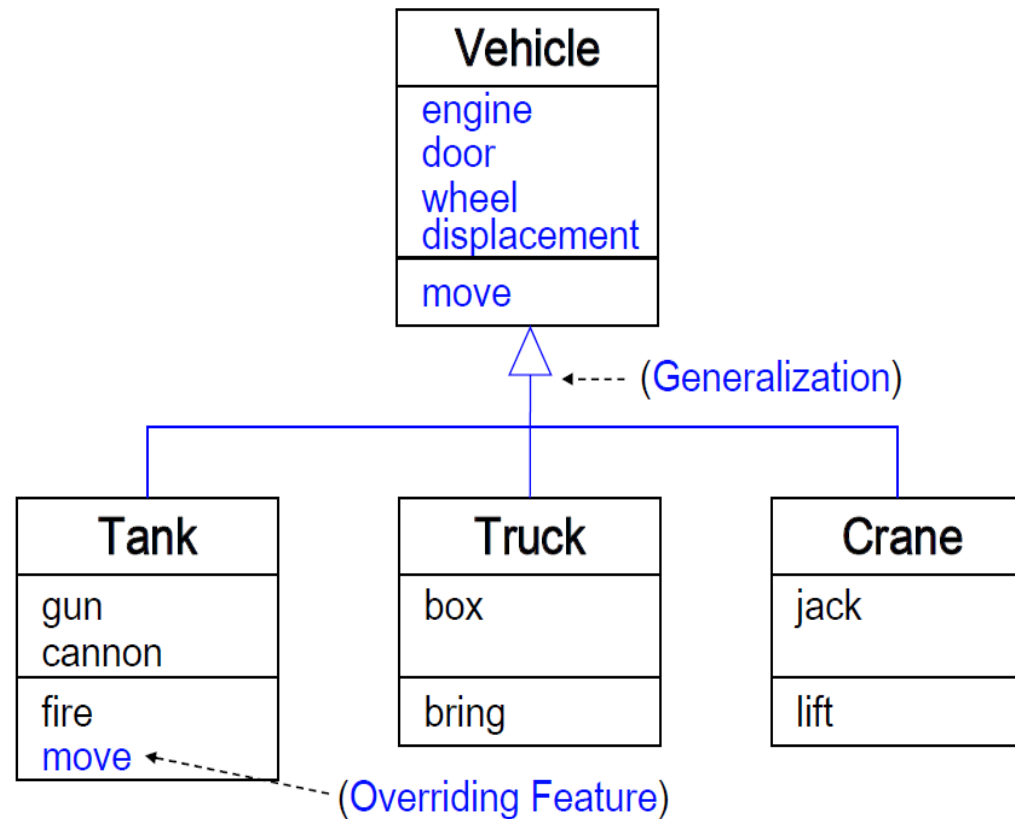
- A large **hollow arrowhead** denotes generalization.
- The arrow head points to the superclass from its subclass.



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Generalization and Inheritance...

For Example:



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2. Aggregation and Composition Relationships

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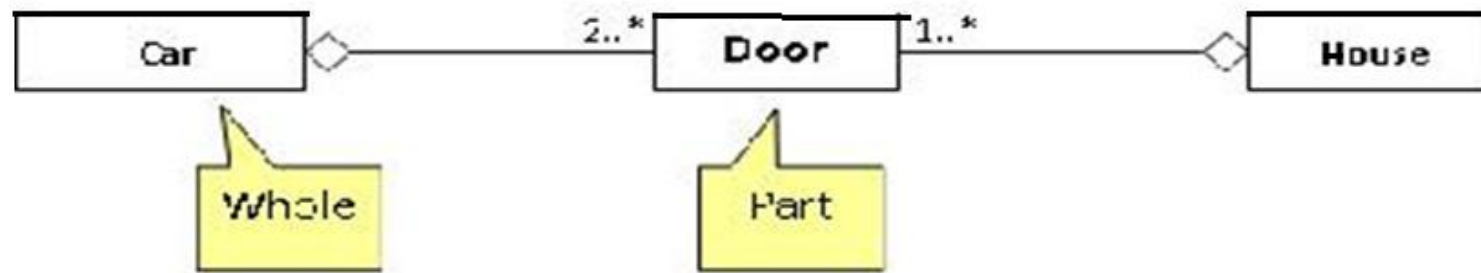
- Aggregation is a **strong form of association** in which **aggregate object is made of constituent parts**.
- If two objects are tightly bound by a **part whole** relationship, it is an aggregation.
- The aggregation is represented by symbol



- For example:



- Aggregation another example:



- Composition is a **form of aggregation** with two additional constraints.
 - A constituent part can belong to at-most one assembly.
 - Composition between two objects, the composed object cannot exist without other object
- The UML notation for composition is



- Example:

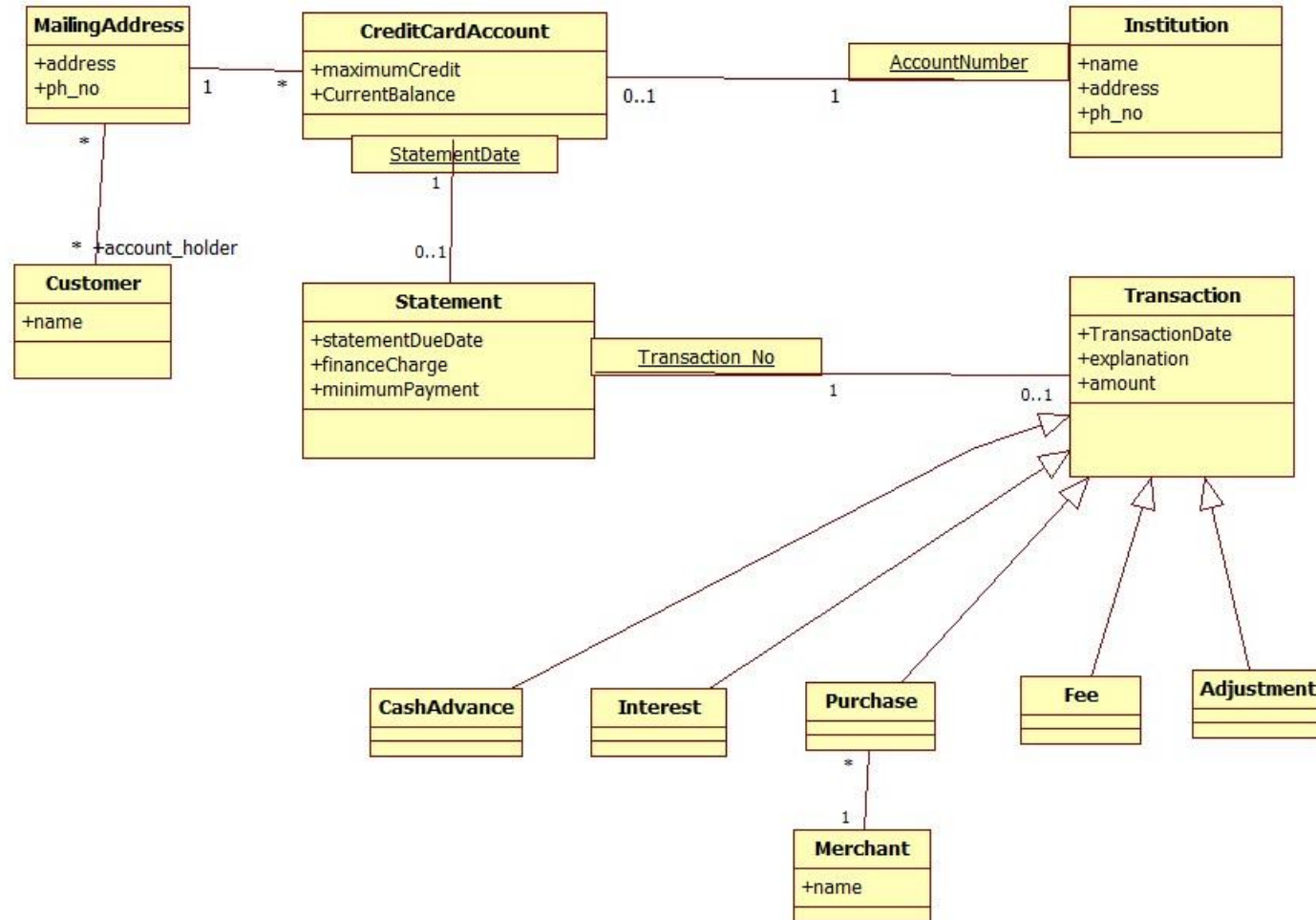


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Class Modeling



A sample class model



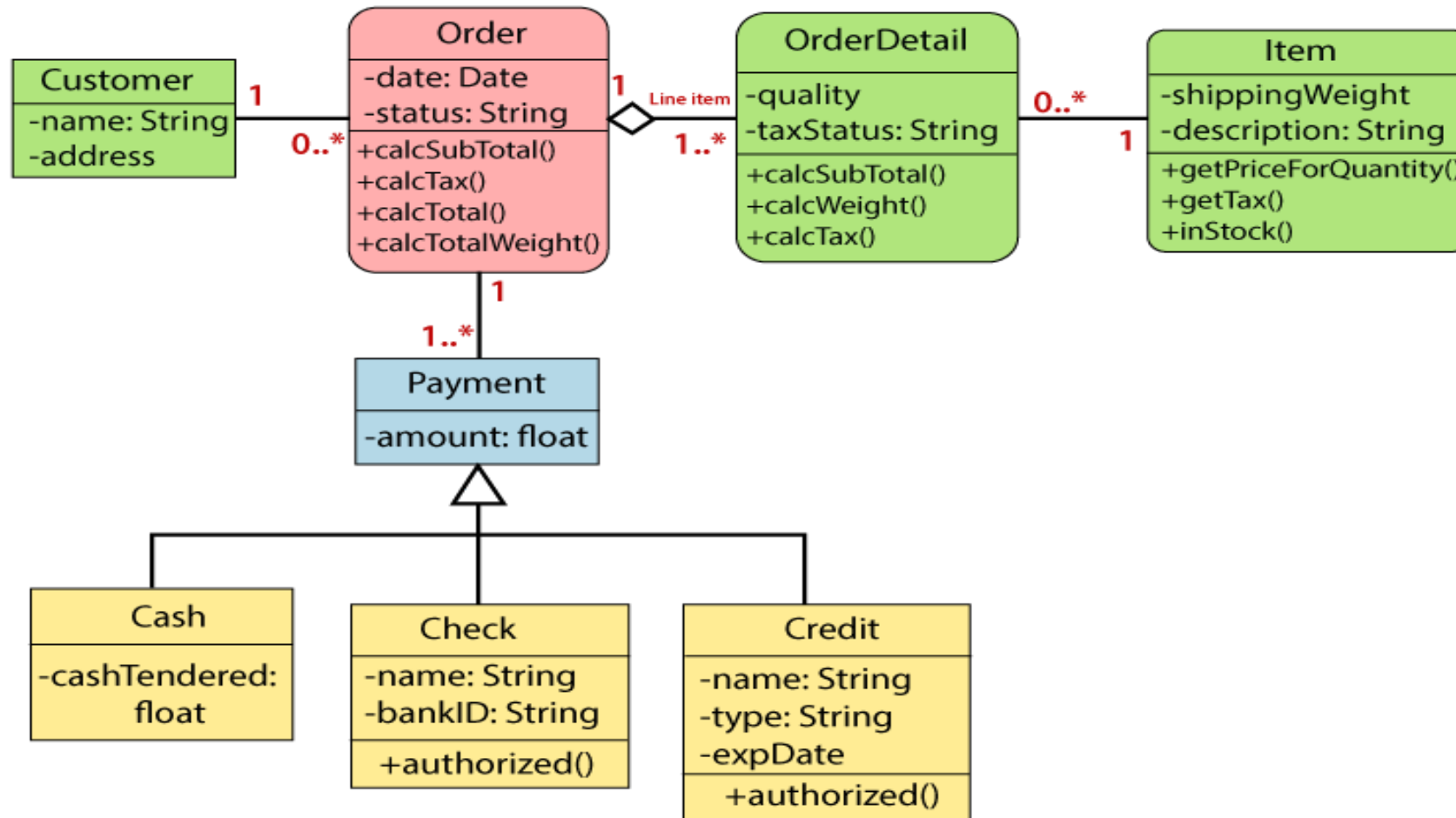
Construct a class model for– **Order Processing system**

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Class Modeling



A class model – Order Processing system





THANK YOU

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